

Python Ultimate Guide

◆ Fundamentals

- └  Variables: `x = 5`
- └  Print: `print("Hello, World!")`
- └  Comments:
 - └ Single-line: `# Comment`
 - └ Multi-line: `'''Comment'''`

◆ Data Types

- └  Primitive:
 - └ String: `"Hello"`
 - └ Integer: `42`
 - └ Float: `3.14`
 - └ Boolean: `True`
- └  Collections:
 - └ List: `[1, 2, 3]`
 - └ Tuple: `(1, 2, 3)`
 - └ Set: `{1, 2, 3}`
 - └ Dictionary: `{"key": "value"}`

◆ Operators

- └  Arithmetic: `+`, `-`, `*`, `/`, `//`, `%`, `**`
- └  Comparison: `==`, `!=`, `<`, `>`, `<=`, `>=`
- └  Logical: `and`, `or`, `not`
- └  Membership: `in`, `not in`
- └  Identity: `is`, `is not`

◆ Conditionals

- └ ✂ If: `if x > y:`
- └ ✂ Elif: `elif x < y:`
- └ ✂ Else: `else:`

◆ Loops

- └ ✂ For: `for x in range(5):`
- └ ✂ While: `while x < 5:`
- └ ✂ Break: `break`
- └ ✂ Continue: `continue`

◆ Functions

- └ ✂ Defining: `def my_function():`
- └ ✂ Calling: `my_function()`
- └ ✂ Default parameters: `def func(x, y=0):`
- └ ✂ Variable-length arguments: `def func(*args, **kwargs):`

◆ Classes & Objects

- └ ✂ Class definition: `class MyClass:`
- └ ✂ Constructor: `def __init__(self):`
- └ ✂ Instance methods: `def method(self):`
- └ ✂ Class variables: `class_var = 0`
- └ ✂ Object instantiation: `my_object = MyClass()`
- └ ✂ Inheritance: `class DerivedClass(BaseClass):`
- └ ✂ Method overriding: `def method(self):`

◆ Error Handling

- └ ✂ Try: `try:`
- └ ✂ Except: `except Exception as e:`
- └ ✂ Raise: `raise ValueError("Error message")`
- └ ✂ Finally: `finally:`

◆ Importing Libraries

- └ ✂ Import: `import numpy`
- └ ✂ Alias: `import numpy as np`
- └ ✂ Specific import: `from math import pi`

◆ File I/O

- └ ✂ Open: with `open("file.txt", "r")` as `file`:
- └ ✂ Read: `file.read()`
- └ ✂ Write: with `open("file.txt", "w")` as `file`:
- └ ✂ Append: with `open("file.txt", "a")` as `file`:

◆ List Comprehensions

- └ ✂ Syntax: `[expression for item in iterable if condition]`

◆ Lambda Functions

- └ ✂ Syntax: `lambda arguments: expression`

◆ Iterators & Generators

- └ ✂ Iterator: `iter(obj)`
- └ ✂ Next item: `next(iterator)`
- └ ✂ Generator function: `def my_generator(): yield value`
- └ ✂ Generator expression: `(expression for item in iterable if condition)`

◆ Context Managers

- └ ✂ Defining: `class MyContext:`
- └ ✂ Enter method: `def __enter__(self):`
- └ ✂ Exit method: `def __exit__(self, exc_type, exc_value, traceback):`
- └ ✂ Using: `with MyContext() as my_context:`

◆ Built-in Functions

- └ ✂ `len(obj)` → Length of object
- └ 📊 `sum(iterable[, start])` → Sum of elements
- └ 📈 `max(iterable[, key])` → Maximum element

- └  min(iterable[, key]) → Minimum element
- └  sorted(iterable[, key][, reverse]) → Sorted list
- └  range(stop[, start][, step]) → Sequence of numbers
- └  zip(*iterables) → Iterator of tuples
- └  map(function, iterable) → Apply function to all items
- └  filter(function, iterable) → Filter elements by function
- └  isinstance(obj, classinfo) → Check object's class

◆ String Methods

- └  lower() → Lowercase
- └  upper() → Uppercase
- └ strip([chars]) → Remove leading/trailing characters
- └  split([sep][, maxsplit]) → Split by separator
- └ ⇄ replace(old, new[, count]) → Replace substring
- └  find(sub[, start][, end]) → Find substring index
- └  format(*args, **kwargs) → Format string

◆ List Methods

- └  append(item) → Add item to end
- └  extend(iterable) → Add elements of iterable
- └  insert(index, item) → Insert item at index
- └  remove(item) → Remove first occurrence
- └  pop([index]) → Remove & return item
- └  index(item[, start][, end]) → Find item index
- └  count(item) → Count occurrences
- └  sort([key][, reverse]) → Sort list
- └  reverse() → Reverse list

◆ Dictionary Methods

- └ 🔑 `keys()` → View list of keys
- └ 📌 `values()` → View list of values
- └ 📁 `items()` → View key-value pairs
- └ 🔍 `get(key[, default])` → Get value for key
- └ 📝 `update([other])` → Update dictionary
- └ 🗑️ `pop(key[, default])` → Remove & return value
- └ 🗑️ `clear()` → Remove all items

◆ Set Methods

- └ ➕ `add(item)` → Add item
- └ 🌐 `update(iterable)` → Add elements of iterable
- └ ✖ `discard(item)` → Remove item if present
- └ 🚫 `remove(item)` → Remove item or raise `KeyError`
- └ 🗑️ `pop()` → Remove & return item
- └ 🗑️ `clear()` → Remove all items
- └ 📁 `union(*others)` → Union of sets
- └ 🔧 `intersection(*others)` → Intersection of sets
- └ = `difference(*others)` → Difference of sets
- └ ⬇ `issubset(other)` → Check if subset
- └ 🌐 `issuperset(other)` → Check if superset

◆ Regular Expressions

- └ 📄 `import re`
- └ 🔍 `re.search(pattern, string)`
- └ ⚓ `re.match(pattern, string)`
- └ 🔍 `re.findall(pattern, string)`
- └ 🔄 `re.sub(pattern, repl, string)`
- └ ✂ `Common patterns:`

- └ \d: Digit
- └ \w: Word character
- └ \s: Whitespace
- └ .: Any character (except newline)
- └ ^: Start of string
- └ \$: End of string
- └ *: Zero or more repetitions
- └ +: One or more repetitions
- └ ?: Zero or one repetition
- └ {n}: Exactly n repetitions
- └ {n,}: At least n repetitions
- └ {,m}: At most m repetitions
- └ {n,m}: Between n and m repetitions (inclusive)

◆ Decorators

- └ 📄 Defining: `def my_decorator(func):`
- └ 📄 Applying: `@my_decorator`

◆ Modules & Packages

- └ 📁 Creating a module: Save as `.py` file
- └ 📄 Importing a module: `import my_module`
- └ 📁 Creating a package: Create directory with `__init__.py`
- └ 📄 Importing from a package: `from my_package import my_module`

◆ Virtual Environments

- └ 🌐 Creating: `python -m venv myenv`
- └ ✨ Activating:
- └ Windows: `myenv\Scripts\activate`
- └ Unix/Mac: `source myenv/bin/activate`
- └ 🚫 Deactivating: `deactivate`

💠 Package Management (pip)

- └ 📄 Install: `pip install package_name`
- └ 📄 Uninstall: `pip uninstall package_name`
- └ 📄 Upgrade: `pip install --upgrade package_name`
- └ 📄 List installed packages: `pip list`
- └ 📄 Show package details: `pip show package_name`

💠 Date & Time

- └ 📄 import `datetime`
- └ 🕒 Current date & time: `datetime.datetime.now()`
- └ 📅 Date object: `datetime.date(year, month, day)`
- └ 🕒 Time object: `datetime.time(hour, minute, second, microsecond)`
- └ 🕒 Format: `datetime.datetime.strftime(format)`
- └ 🕒 Parse: `datetime.datetime.strptime(date_string, format)`
- └ 📅 Common format codes: `%Y, %m, %d, %H, %M, %S`

💠 JSON

- └ 📄 import `json`
- └ 🕒 JSON to Python: `json.loads(json_string)`
- └ 🕒 Python to JSON: `json.dumps(obj)`
- └ 📄 Read from file: `json.load(file)`
- └ 📄 Write to file: `json.dump(obj, file)`

💠 Threading

- └ 📄 import `threading`
- └ 🕒 Create a thread: `t = threading.Thread(target=function, args=(arg1, arg2))`
- └ 🚀 Start a thread: `t.start()`
- └ 📄 Wait for thread to finish: `t.join()`

◆ Multiprocessing

└  `import multiprocessing`

└  Create a process: `p = multiprocessing.Process(target=function, args=(arg1, arg2))`

└  Start a process: `p.start()`

└  Wait for process to finish: `p.join()`

◆ Working with Databases (SQLite)

└  `import sqlite3`

└  Connect to a database: `conn = sqlite3.connect('mydb.sqlite')`

└  Cursor object: `cursor = conn.cursor()`

└  Execute SQL commands: `cursor.execute("CREATE TABLE my_table (id INTEGER, name TEXT)")`

└  Commit changes: `conn.commit()`

└  Fetch results: `cursor.fetchall()`

└  Close the connection: `conn.close()`

◆ Web Scraping (BeautifulSoup)

└  `from bs4 import BeautifulSoup`

└  Create a BeautifulSoup object: `soup = BeautifulSoup(html_content, 'html.parser')`

└  Find elements by tag: `soup.find_all('tag_name')`

└  Access element attributes: `element['attribute_name']`

└  Get element text: `element.text`

💠 Web Requests (Requests)

- └ 📡 import requests
- └ 📄 GET request: response = requests.get(url)
- └ 📡 POST request: response = requests.post(url, data=payload)
- └ 📄 Response content: response.content
- └ 🌿 JSON response: response.json()
- └ 🚦 Response status code: response.status_code

💠 Web Development (Flask)

- └ 📡 from flask import Flask, render_template, request, redirect, url_for
- └ 🌐 Create a Flask app: app = Flask(__name__)
- └ 🚩 Define a route: @app.route('/path', methods=['GET', 'POST'])
- └ 🚶 Run the app: app.run(debug=True)
- └ 📄 Return a response: return "Hello, World!"
- └ 🎨 Render a template: return render_template('template.html', variable=value)
- └ 📄 Access request data: request.form['input_name']
- └ 🔄 Redirect to another route: return redirect(url_for('route_function'))

💠 Data Science Libraries

- └ 🖋️ NumPy: import numpy as np
- └ 📊 pandas: import pandas as pd
- └ 🎨 Matplotlib: import matplotlib.pyplot as plt
- └ 📈 seaborn: import seaborn as sns
- └ 🤖 scikit-learn: import sklearn
- └ 🧠 TensorFlow: import tensorflow as tf
- └ 🚀 Keras: from tensorflow import keras
- └ 🔥 PyTorch: import torch

◆ Command Line Arguments (argparse)

└  import argparse

└  Create an ArgumentParser: parser =
argparse.ArgumentParser(description='Description of your program')

└  Add arguments: parser.add_argument('--arg_name', type=str,
help='Description of the argument')

└  Parse arguments: args = parser.parse_args()

└  Access argument values: args.arg_name

◆ Logging

└  import logging

└  Basic configuration: logging.basicConfig(level=logging.DEBUG,
format='%(asctime)s - %(levelname)s - %(message)s')

└  Logging levels: logging.debug(), logging.info(), logging.warning(),
logging.error(), logging.critical()

◆ Environment Variables

└  import os

└  Get an environment variable: os.environ.get('VAR_NAME')

└  Set an environment variable: os.environ['VAR_NAME'] = 'value'

◆ Type Hints

└  from typing import List, Dict, Tuple, Optional, Union, Any

└  Function type hints: def my_function(param: int, optional_param:
Optional[str] = None) -> List[int]:

└  Variable type hints: my_variable: Dict[str, int] = {}